Detecting Adverse Drug Events: A Comparative NLP Approach

Evaluating Traditional Machine Learning vs. Transformers in Healthcare Pharmacovigilance

Niharika Kunapuli | DS & AI Capstone

Problem Definition & Impact

The Need: The FDA relies on manufacturers, doctors, and patients to voluntarily report Adverse Drug Events (ADEs). This leads to under-reporting issues and a slow reaction time by the FDA.

Research Question: Can we automate the extraction of negative experiences and ADEs from unstructured text using NLP?

Real-world Impact: Automating this pipeline creates an early-warning surveillance system, potentially saving lives by catching complications faster.

Dataset & Feature Engineering

Dataset & Targets

Utilized the UCI Machine Learning Drug Review Dataset (~3,100 records). Created a binary classification system based on patient ratings: Positive/No ADE vs. Negative/ADE.

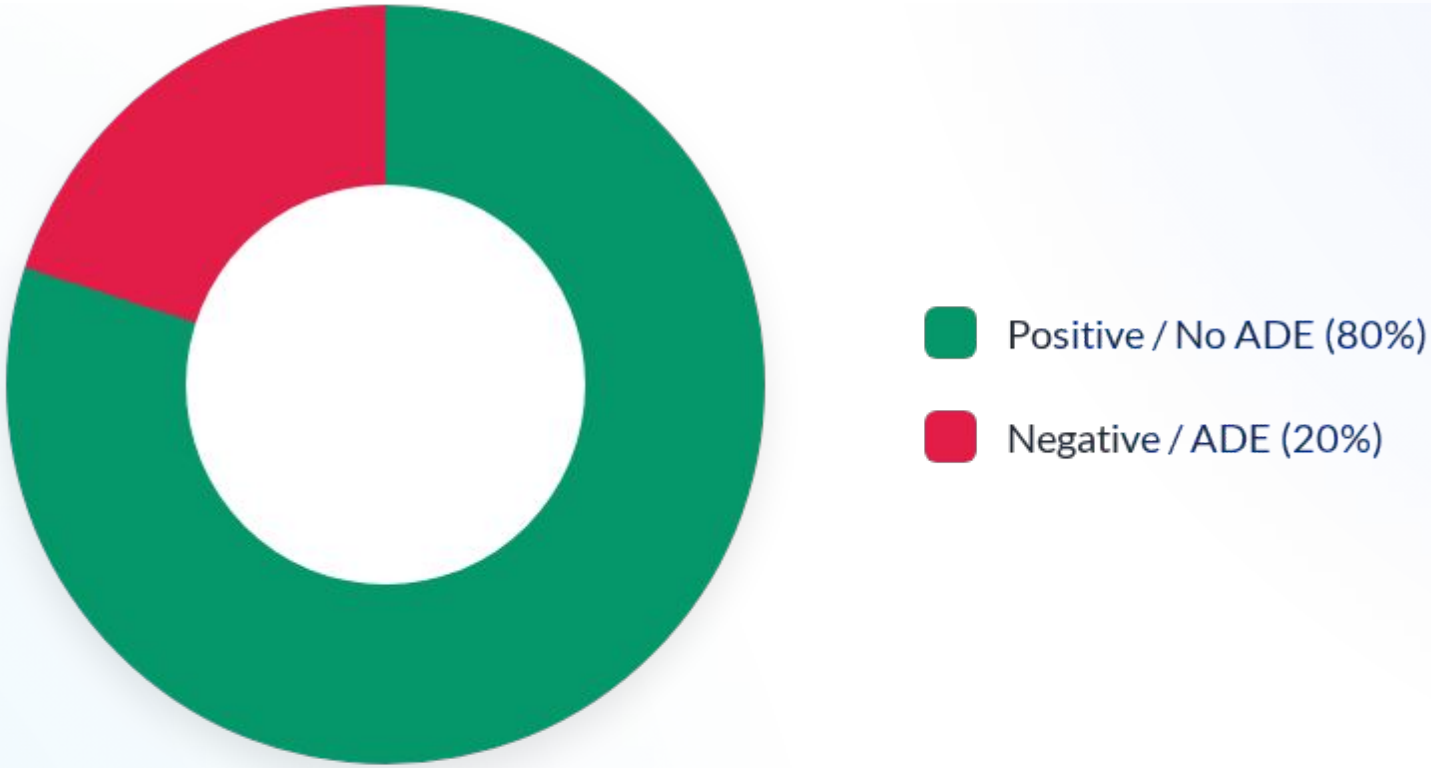
Data Cleaning

Ensured high-quality input by dropping missing targets (NaNs), scrubbing HTML artifacts from the raw text, and systematically handling whitespace anomalies.

Feature Engineering

Applied a "Prefixing Hack": Concatenated separate fields into one string with structural prefixes (e.g., BENEFITS: ... SIDE_EFFECTS: ...) to guide model comprehension.

Class Imbalance & NLP Analysis



Key Insight: The heavy class imbalance (mostly positive reviews) dictates the critical need for specialized loss functions.

Model Choice & Architecture

The Baseline: TF-IDF + LogReg

Traditional Term Frequency-Inverse Document Frequency Vectorization paired with Logistic Regression.

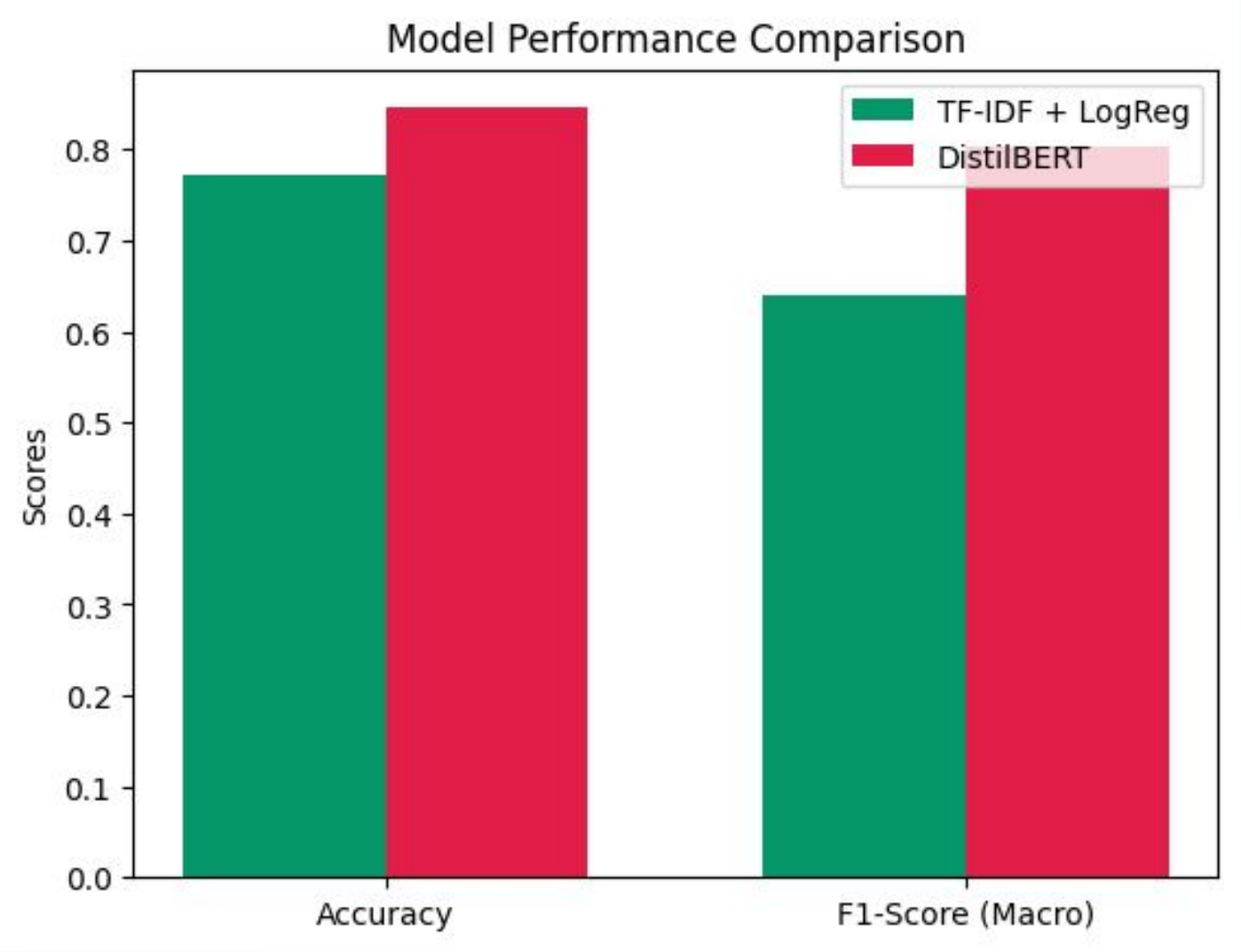
Why: Highly interpretable, computationally cheap, and mathematically optimized for high-dimensional sparse matrices. A strong, reliable starting point.

The Advanced Approach: DistilBERT

Hugging Face Transformer model offering a modern, deep-learning approach to natural language processing.

Why: Powerful self-attention mechanisms can capture complex contextual negations (e.g., "did not cause severe pain") that traditionally trip up bag-of-words models.

Comparing Initial Model Performance



Error Analysis: Why Context Matters

Review Excerpt (Row 31)	TF-IDF Predict	DistilBERT Predict	True Label
"BENEFITS: Sore throat considerably better... SIDE_EFFECTS: Extreme nausea and continuous vomiting."	Positive / No ADE	Negative / ADE	Negative / ADE
"BENEFITS: Cleared up the rash entirely... SIDE_EFFECTS: Did not experience any severe headaches as warned."	Negative / ADE	Positive / No ADE	Positive / No ADE

Key Insight: While overall metrics favor TF-IDF's minority recall, DistilBERT is vastly superior at understanding complex semantics—such as recognizing when a severe side effect overrides a minor benefit, or processing contextual negations.

Execution: The TF-IDF Baseline

80%

Validation Accuracy

WEIGHTED METRICS	CLASS 0 (ADE)	CLASS 1 (NO ADE)
Precision	0.65	0.87
Recall	0.70	0.84
F1-Score	0.67	0.85

Macro Avg F1: 0.76

Methodology & Results

- Applied ngram_range=(1,2) to capture essential bigrams representing basic context.
- Utilized class_weight='balanced' to force the model to respect the minority ADE class.
- **Interpretation:** The baseline performed surprisingly well. It effectively identified adverse events once the minority class was heavily weighted.

Execution: The DistilBERT Model

84%

Validation Accuracy

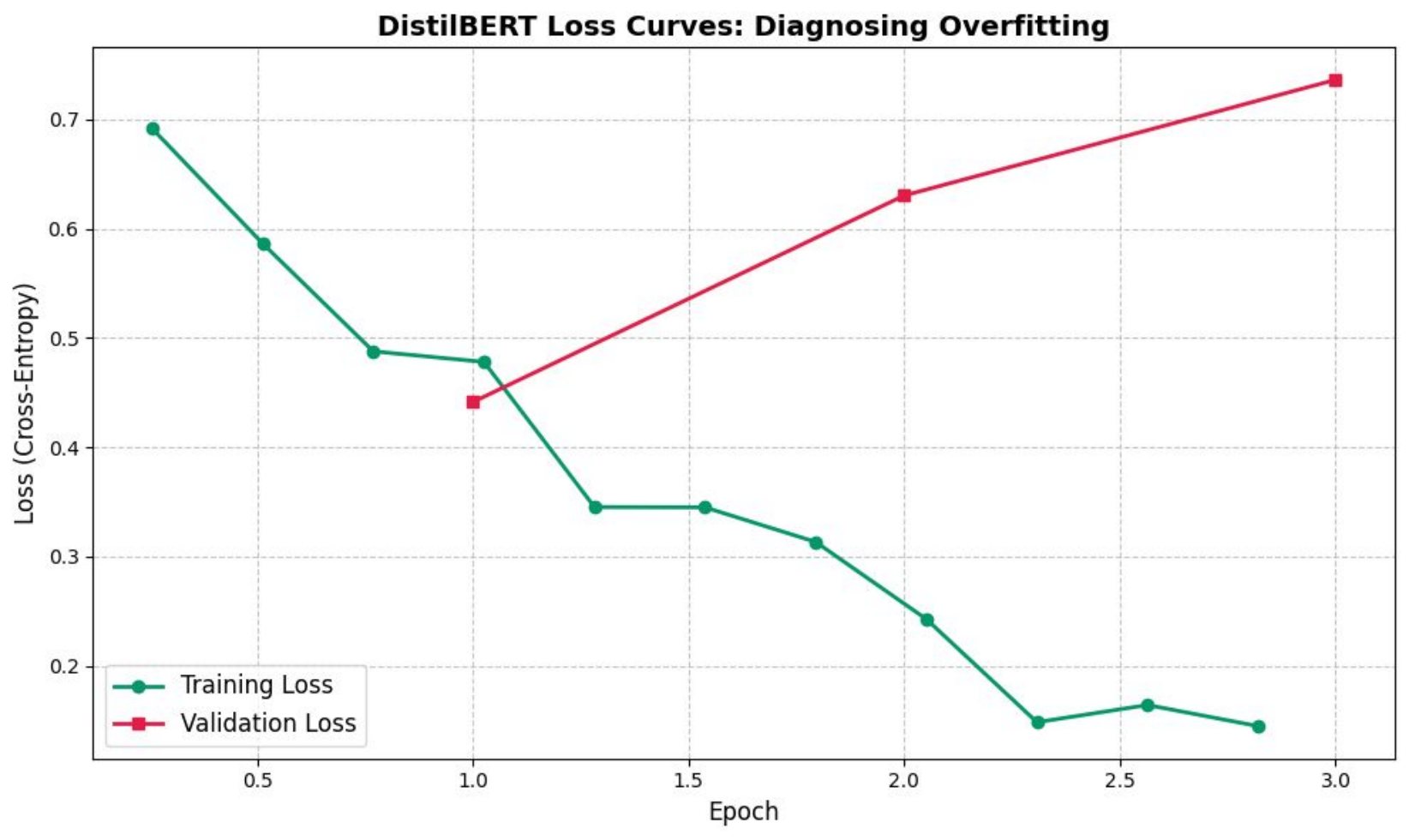
WEIGHTED METRICS	CLASS 0 (ADE)	CLASS 1 (NO ADE)
Precision	0.76	0.87
Recall	0.68	0.91
F1-Score	0.72	0.89

Macro Avg F1: 0.80

Methodology & Results

- Fine-tuned over 3 epochs on a T4 GPU infrastructure.
- Implemented a custom subclassed Trainer to inject a weighted Cross-Entropy loss function, directly combating class imbalance.
- **Interpretation:** Higher overall accuracy, but the Transformer struggled to beat the heavily tuned traditional baseline on minority class recall.

Diagnosing Overfitting Limits



Result Interpretation: The divergence in the curves explains why Transformer metrics stalled. The 66M-parameter model is rapidly overfitting to the small dataset (~2,400 rows), memorizing text instead of generalizing.

Conclusion & Trade-off

- Deep learning is bottlenecked by data volume
- For datasets of this size, computationally cheap Logistic Regression remains the safer, more robust deployment choice
- In production, TF-IDF acts as the rapid primary inference engine
- Transformers require a massive data-gathering phase before deployment in critical pharmacovigilance